Experiment tracking and model registry

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Recap: Main problem → ML Model pipeline

obj: Build a model → perf of model on Train / Test → Near ≤ 5%

The performance should be reasonably good

Encounter

perf on both train & test data are low.

Model → Train perf ≈ 35% ; Test data ≈ 30%

Underfitting → why?

Training → During Model Building

The parameter Chosen for a ML algo.

KNN ⇒ n_neighbors = 75. ← Config.

Model simple

Less scope for learning

Under performance

Model → Train data Good 95% ; Test data → poor 65%

Overfitting → why.

Training →

The param Choosen for an ML algo

Complex param

KNN n_neigh = 1 → Complex

Model Complex

Infinite scope for lear

↓
Under performance

Infinite scope for lear
↓
Model learns NOISE
(Memorizes)
Train data
↓
Performs very well on Train data

But under perform → Test data

How to address overfitting/ underfitting problems?

Naive: for every Config of ML algo

KNN. (4)

n_neigh = 1
= 2
= 3

	Train	Perf	Test	Perf
OV	✓	G	✓	P
Good	✓	G	✓	G
Und	✓	P	✓	P

KNN

n_neigh	Train	Test
1	80	50
2	85	55
3	85	60
4	85	65
5	90	70
⋮		
15	95	93

as we inc n_neigh ↓ rate at which train perf inc large

→ easy to understand the Char of Test

Test_perf ∝ n_neigh

↘ Know info Test data

⇒ Data Leakage

Cross Validation:

Dataset → Train, Test

CV

Train

n equal parts (5) - (10)

n_split = 5.

Train

CV ⇒

n equal parts (5) - (10)

n_splits = 5.

5 valid.

	P_1	P_2	P_3	P_4	P_5
R_1	Train $P_1 - P_4$	Valid P_5.			
R_2	$P_1, 2, 3, 5$	P_4.			
R_3	$P_1, 2, 4, 5$	P_3.			
		P_2			
		P_1.			

⇒ 5 valid splits

⇒ Train Sc1 Valid Sc1

5 Runs.

	Train	Valid
	T_{Sc2}	V_{Sc2}
	T_{Sc3}	V_{Sc3}
	⋮	⋮
	T_{Sc5}	V_{Sc5}.
	MTs	MVs

@ part Combs → CV

MTs ≈ MVsc reasonably good Conf

go with another Combrg.